



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH	IIT Kharagpur	7.46 / 10
2025	Agricultural Engineering	Anand Agricultural University	8.049 / 10
2021	Intermediate Annual Exam	BSEB	83.8%
2019	Secondary School Examination	CBSE	89.4%

COURSEWORK INFORMATION

GenAI & Agentic Tools for Business Transformation | AI Applications in Agriculture | Probabilistic Risk Assessment | Water Resources System Analysis | Geo-informatics for Land & Water Resources | Advanced Groundwater & Surface Hydrology | Statistical and Machine Learning Methods in Water Resources Engineering | Digital Soil Mapping

SKILLS AND EXPERTISE

Data Science & Generative AI: Python, R, TensorFlow, Scikit-learn, LangChain, RAG, Transformers, LLMs, XGBoost, Hypothesis Testing.
Geospatial Analysis: Google Earth Engine, GeoPandas, PostGIS, QGIS, Rasterio, Xarray, GDAL.
Tools, Cloud & Visualization: AWS, Docker, Git, Streamlit, MLFlow, Plotly, Matplotlib, Folium.

INTERNSHIPS

Research Intern | Sardar Patel Renewable Energy Research Institute (SPRERI) | Gujarat | Aug 2024 - Nov 2024

- Applied **Central Composite Design (CCD)** in Design-Expert to model **non-linear optical responses**, optimizing nanofluid stability parameters
- Analyzed **time-series thermal data**, identifying **10% nanofluid concentration** as optimal for solar heat-retention efficiency
- Analyzed **DSC thermodynamic phase-change data** of NaNO₃/KNO₂ mixtures, isolating the **60:40 ratio** with **92.79 J/g latent heat capacity**

Summer Intern | Swaraj Division, Mahindra & Mahindra Ltd. | Jharkhand | Jun 2024

- Performed **data-driven market and product segmentation analysis** using sales, enquiry, and regional datasets to identify demand patterns
- Conducted **financial and performance analysis** of sales and service operations, extracting actionable business insights
- Analyzed **dealer management and after-sales service data** to support operational efficiency and decision-making

PROJECTS

Bihar Hydro-Climatic Risk Atlas | Full-Stack Geospatial Data Science Project

- Built a **Climate-AI decision system** using **Random Forest ML ($R^2 > 0.85$)** on multi-source satellite data (CHIRPS, MODIS, Sentinel-2) to model **flood pressure and groundwater stress** at block scale (2019–2024)
- Engineered **spatio-temporal climate features** (rainfall anomalies, NDVI, ET/Rain interaction) and applied **SHAP explainability** to uncover non-linear tipping points driving hydro-climatic risk
- Developed a **composite risk index + trend degradation model (OLS)** and deployed an **interactive Streamlit app** enabling data-driven prioritization of climate adaptation interventions

Study of Different Thermal Drying Material for Solar Drying Application | Major B.Tech Project

- Designed and instrumented an **indirect passive solar dryer**, generating **high-resolution time-series datasets** (10-min intervals) for temperature, airflow, solar irradiance, drying rate, and efficiency across multiple **TES configurations**
- Performed **comparative statistical analysis and performance benchmarking** of TES materials (pebbles, grit, recycled metal bottle caps), identifying **metal bottle caps as the optimal TES** with **59.7% thermal efficiency** and **161% heat recovery ratio**
- Analyzed **drying kinetics and quality metrics** across slice thicknesses using moisture models and proximate/nutritional data, deriving **data-backed optimization insights** for energy-efficient, climate-resilient food processing

IoT-Enabled Automated Fertigation System | Precision Agriculture

- Built an **IoT-based precision fertigation system** leveraging **soil moisture and nutrient sensor data** to enable real-time, closed-loop irrigation control at **40–55% VWC**
- Developed **edge decision logic (Arduino)** for automated water–nutrient dosing, translating sensor signals into efficient actuation
- Delivered **38.5% reduction in water consumption** versus flood irrigation (1.7 L vs 2.8 L), showcasing **measurable climate and resource efficiency impact**

Ultrasonic Guided Bluetooth-Controlled Spraying Rover | Minor Project

- Engineered a **semi-autonomous spraying rover** using **ultrasonic sensing, Bluetooth control, and Arduino Uno**, achieving **192 ms response latency** and **94% obstacle detection accuracy** via optimized control logic
- Benchmarked field performance at **2,500 m²/hr coverage** and **25/hr operating cost**, enabling scalable, low-cost deployment
- Reduced **operator chemical exposure by >90%**, delivering measurable **health, safety, and sustainability impact**

PUBLICATIONS

Design, Implementation, and Performance Assessment of an Ultrasonic-Guided Bluetooth-Controlled Spraying Rover for Vegetable Crop Rows **Ankit Kumar**, S. K. Jogunuri, D. K. Vyas Journal of AgriSearch, 12(4): 237-244, December 2025 DOI: 10.21921/jas.v12i04.15262

COMPETITION/CONFERENCE

58th ISAE Annual Convention & International Symposium | Parbhani, Maharashtra

Presented research paper on **advanced thermal energy storage technologies** at the **58th ISAE Annual Convention (Nov 2024)**, covering **TES materials, applications, and sustainability challenges**

35th National Convention of Agricultural Engineers (IEI) | JNKVV, Jabalpur

Presented Paper: “*Design and Development of Automated Fertigation System using IoT*” - developed an **IoT-based fertigation system** maintaining **40–55% VWC** with **~38.5% water savings** over flood irrigation.

AWARDS AND ACHIEVEMENTS

University Gold Medalist: Secured highest B.Tech GPA at Anand Agricultural University.
GATE Top Ranker: Achieved AIR 69 (2025) and AIR 111 (2024) in Agricultural Engineering.